

DEDEKIND CUBICAL SETS DO NOT PRESENT SPACES

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ABSTRACT.

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1. INTRODUCTION

Cubical models of homotopy type theory interpret the type-theoretic judgments in presheaves over a cube category, and carry *cubical-type model structures* whose fibrant objects are the uniform Kan complexes [5, 4]. Buchholtz and Sattler determined in 2018 that almost all of the concrete cubical-type model structures present $(\infty, 1)$ -categories inequivalent to ∞ -groupoids; the single exception left open was the Sattler model structure on presheaves over the *Dedekind cube category* $\square_{\wedge\vee}$ — the Lawvere theory of bounded distributive lattices, i.e. cubes with cartesian structure and both connections [2]. Cavallo and Sattler recently proved that with exactly *one* connection the cubical-type model structure is Quillen equivalent to spaces, and showed at the same time that their method — relative elegance — cannot apply to $\square_{\wedge\vee}$: the Dedekind cube category admits no relatively elegant embedding into any Reedy category, because monotone circles (crown posets) inside cubes support self-coverings of arbitrary winding number [2].

This paper settles the remaining case, negatively.

Theorem 1.1. *On the Dedekind cube category, the cubical-type weak equivalences differ from the test weak equivalences: $W_{\text{ty}} \neq W_{\text{test}}$. In particular the cubical-type model structure on Dedekind cubical sets does not present the homotopy theory of spaces.*

The witness is disarmingly small.

Definition 1.2 (The monotone dunce hat). Let $W_{\wedge\vee} := \square^2 / ((0, u) \sim (u, 0), (u, u) \sim (u, 1))$: the quotient of the representable square by the presheaf congruence generated by identifying the left edge with the bottom edge, and the diagonal with the top edge.

Theorem 1.3. *$W_{\wedge\vee}$ is test-contractible, but its identity map is not homotopic to a constant map in the cubical-type structure. Hence the quotient map or the constant map of $W_{\wedge\vee}$ realizes the difference $W_{\text{ty}} \neq W_{\text{test}}$.*

The proof of type-side rigidity is a *provenance analysis*: the fibrant replacement of $W_{\wedge\vee}$ is a transfinite composition of pushouts of open-box inclusions, so each cell of it is either a cell of $W_{\wedge\vee}$ or a fresh cell of a uniquely determined filling stage; a hypothetical contracting cylinder is analyzed by where its structural restrictions live. Three mechanisms close all cases:

- the finitely many cylinders whose cells lie in $W_{\wedge\vee}$ itself are enumerated exactly (twice, by independent implementations): only the constant cylinder starts at the identity;
- a fresh cylinder's *diagonal track* can never attach to earlier stages, by the following lemma applied to its components — the geometric heart of the paper;
- the remaining exact equations force, at the identity end of the cylinder, a constant edge where $W_{\wedge\vee}$ has none.

Lemma 1.4 (Diagonal-track lemma). *Let $\varphi: \{0, 1\}^3 \rightarrow \{0, 1\}$ be monotone. If φ is constant on the diagonal cylinder $\{(u, u, t)\}$, then φ is constant.*

Proof. The diagonal cylinder contains both the bottom vertex $(0, 0, 0)$ and the top vertex $(1, 1, 1)$ of the cube. A monotone function taking value e at the top satisfies $\varphi \leq e$ everywhere, and value e at the bottom forces $\varphi \geq e$ everywhere. $\square$

No other track has this property — the middle variable v is constant on the top cylinder $\{(u, 1, t)\}$ but not constant — and this asymmetry is precisely why the *diagonal* fold in $W_{\wedge\vee}$ is fatal. The two folds of $W_{\wedge\vee}$ cooperate: the swap fold $(0, u) \sim (u, 0)$ blocks the $\vee$ -connection contraction of the square, the diagonal fold $(u, u) \sim (u, 1)$ blocks the $\wedge$ -connection contraction, and Lemma 1.4 shows that no amount of higher-dimensional filling can reassemble a contraction. Each fold alone is harmless: both single-fold quotients are strictly contractible.

Context. The phase diagram of cubical sites is now complete. Writing “=” for agreement of the cubical-type and test weak equivalences:

site	status	reference
cartesian (no connections)	$\neq$	[1] (equivariant repair =)
one connection	$=$	[2]
Dedekind (two connections)	$\neq$	this paper
De Morgan, Kleene, Boolean (reversal)	$\neq$	[6, 7, 8]

Exactly one standard cubical site presents spaces. On the reversal sites the failure is governed by reflection subgroups [7, 8]; here it is governed by the interaction of diagonals with connections — the same interaction that destroys Eilenberg–Zilber decompositions [2] — and the witness is not a group quotient but a two-relation congruence quotient.

2. PRELIMINARIES

$\square_{\wedge\vee}$ has objects $[n]$, $n \geq 0$, and morphisms $\square_{\wedge\vee}([m], [n]) = D(m)^n$, where $D(m)$ is the free bounded distributive lattice on m generators, realized as the lattice of monotone maps $\{0, 1\}^m \rightarrow \{0, 1\}$; composition is substitution. Presheaves are *Dedekind cubical sets*; a cell of X at level m is an element of $X([m])$, and for the representable $\square^n$ a level- m cell is an n -tuple over $D(m)$. The objects $[a] \times [b] = [a + b]$ are products, so $\square^a \times \square^b \cong \square^{a+b}$.

The *cubical-type model structure* has cofibrations the monomorphisms and fibrant objects the uniform Kan complexes; its generating trivial cofibrations are the open-box inclusions $\square_{i,*}^m \hookrightarrow \square^m$ ($m \geq 1$), and the fibrant replacement $R(X)$ of any X is a transfinite composition of pushouts of such inclusions [5, 4]. We use three standard consequences.

Lemma 2.1 (Provenance). *Let $X = Y_0 \subseteq Y_1 \subseteq \cdots \subseteq Y$ be a transfinite composition of pushouts of open-box inclusions $\square_\beta \hookrightarrow \square^{m_\beta}$ along attaching*

maps $\sqcap_\beta \rightarrow Y_{<\beta}$. Then every cell of Y is either a cell of X (old) or arises, for a unique stage β and a unique cell c of $\square^{m_\beta}$ not lying in $\sqcap_\beta$, as the image of c (fresh, with provenance (β, c)); no two distinct fresh cells and no fresh and old cell are equal, and a fresh cell's restriction along a cube map is computed by restricting c : it stays fresh if the restriction avoids $\sqcap_\beta$, and attaches to the earlier cell given by the attaching map otherwise.

Proof. Pushouts along monomorphisms of presheaves are computed levelwise by disjoint union of the complement, and monomorphisms are stable under pushout and transfinite composition. $\square$

Lemma 2.2 (Maps and cylinders from a quotient). *Let $W = \square^2/(A_1 \sim B_1, A_2 \sim B_2)$ be the coequalizer of two pairs of parallel cells $A_i, B_i \in \square^2([1])$, and Y any presheaf. Then maps $W \rightarrow Y$ are exactly the cells $h \in Y([2])$ with $h \cdot A_i = h \cdot B_i$ ($i = 1, 2$), and maps $W \times \square^1 \rightarrow Y$ are exactly the cells $H \in Y([3])$ with $H \cdot (A_i \times \text{id}) = H \cdot (B_i \times \text{id})$, where $\cdot$ denotes restriction along the evident cube maps. The two ends of such a cylinder are its restrictions at $t = 0, 1$.*

Proof. The universal property of the coequalizer, using $\square^2 = y[2]$, $\square^2 \times \square^1 = y[3]$, and that colimits of presheaves are computed levelwise. $\square$

Lemma 2.3 (Homotopy). *Two maps $f, g: W \rightarrow R(W)$ are homotopic in the cubical-type structure if and only if they are connected by a finite chain of cylinders $W \times \square^1 \rightarrow R(W)$ as in Lemma 2.2. The identity class of W is the class of the composite $\iota: W \rightarrow R(W)$ of the identity with the replacement unit, and W is type-contractible iff ι is homotopic to a constant map.*

Proof. All objects are cofibrant (cofibrations are the monomorphisms), $R(W)$ is fibrant, and $W \times \square^1$ is a cylinder object: the end inclusions are monomorphisms and the projection is a weak equivalence because the interval, having connections, is contractible. Left homotopy over a cylinder object between a cofibrant object and a fibrant one is an equivalence relation computed by finite chains. $\square$

3. THE IDENTITY IS ISOLATED

Fix $W = W_{\wedge \vee}$, with the two generating pairs $A_1 = (0, u)$, $B_1 = (u, 0)$ (swap fold) and $A_2 = (u, u)$, $B_2 = (u, 1)$ (diagonal fold), and let $Y = R(W)$ with its filling presentation as in Lemma 2.1. For a cylinder $H \in Y([3])$ write $d_L := H \cdot (A_2 \times \text{id})$, $d_R := H \cdot (B_2 \times \text{id})$ for its diagonal and top tracks.

Two finite facts about W itself enter the proof; both are verified exactly, by two independent implementations (Section 5).

Lemma 3.1 (Base facts). *(i) The class of the generic cell of W at level 2 is a singleton: no other cell of $\square^2$ is congruent to (x, y) . (ii) Among the cylinders $H \in W([3])$ (old cells), the only one with an end in the identity class is the constant cylinder. (iii) The class of the left edge $[(0, v)]$ of W is not the class of a constant edge.*

Proof. (i) The congruence is generated by the cells $A_i \cdot u \sim B_i \cdot u$; each generator has a component which is constant or a substitution instance of a single variable in a way the generic cell is not, and the transitive closure preserves this. (ii, iii) Finite enumeration over $D(3)^2$ and $D(1)^2$; artifact, two implementations. $\square$

Theorem 3.2 (Isolation). *Every cylinder $H \in Y([3])$ satisfying the two conditions of Lemma 2.2 with one end in the class of ι has both ends in the class of ι ; indeed the only such cylinders are constant in the sense that both ends equal ι . Consequently ι is not homotopic to a constant map, and W is not type-contractible.*

Proof. If H is old (a cell of W), Lemma 3.1(ii) applies. So let H be fresh with provenance (β, c) , c a level-3 cell of $\square^m$ avoiding $\sqcap_\beta$, and suppose the end $e_0 := c \cdot (t=)$ has class ι .

The diagonal track is fresh. The components of $d_L = c \cdot (A_2 \times \text{id})$ are the monotone functions $c_a(u, u, t)$. If d_L attached, some component would be constant at a present face's value; by Lemma 1.4 that component of c is globally constant, so c itself lies in that present face of $\sqcap_\beta$ — contradicting freshness.

Hence d_R is fresh and $d_L = d_R$ exactly. A fresh cell never equals an old one (Lemma 2.1), so the condition $d_L = d_R$ in Y forces d_R fresh of the same stage and the equality of the underlying cube cells: $c \cdot (A_2 \times \text{id}) = c \cdot (B_2 \times \text{id})$ in $\square^m$.

The end attaches, exactly onto the generic cell. If e_0 were fresh its class would be a fresh singleton, not ι . So e_0 lies in $\sqcap_\beta$: writing f_0 for a present face containing it, $e_0 = f_0 \circ \rho$ for a unique level-2 cell ρ of $\square^{m-1}$, and its attachment is $A \circ \rho$ where $A \in Y_{<\beta}([m-1])$ is the attaching cell of the face. By Lemma 3.1(i) the class ι is a singleton, so $A \circ \rho = \iota$ exactly.

The contradiction. Restrict the exact equation $c \cdot (A_2 \times \text{id}) = c \cdot (B_2 \times \text{id})$ at $t =:$ $e_0 \cdot A_2 = e_0 \cdot B_2$, i.e. $f_0 \circ (\rho \cdot A_2) = f_0 \circ (\rho \cdot B_2)$, and since the face inclusion is monic, $\rho(u, u) = \rho(u, 1)$ as cells of $\square^{m-1}$. Evaluating at $u = 0$ gives the vertex equation $\rho(0, 0) = \rho(0, 1)$. A level-1 cell of a representable cube whose two vertices agree is constant (its components lie in $D(1)$ and are determined by their endpoint values). Hence the left edge $\rho \cdot (0, v)$ of ρ is a constant cell, and applying A ,

$$\iota \cdot (0, v) = A \circ \rho \cdot (0, v) = A \circ (\text{constant}) = \text{a constant edge of } Y_{<\beta},$$

contradicting Lemma 3.1(iii): the left edge of $W_{\wedge \vee}$ is not constant. So no fresh cylinder has an end of class ι , and the theorem follows by transfinite induction over the filling presentation. $\square$

Remark 3.3. The proof uses only the diagonal-fold condition to pin the cylinder and the swap fold to make the base facts true; both folds are needed — each single-fold quotient is strictly contractible (artifact). The theorem holds verbatim for any fibrant replacement built from open-box fillings, of boxes of arbitrary dimension: no dimension bound is used.

4. TEST-CONTRACTIBILITY

We prove test-contractibility by exhibiting $W_{\wedge\vee}$ as an iterated homotopy pushout of contractible presheaves. Recall that in the test structure every object is cofibrant (cofibrations are the monomorphisms) and representables are contractible, so a pushout along a monomorphism is a homotopy pushout, and a wedge of two representable intervals at a vertex is contractible.

Lemma 4.1 (Wedge monomorphisms). *(i) The wedge of the left and bottom edges at their common vertex v_{00} maps monomorphically into $\square^2$. (ii) Let $W_1 := \square^2 / ((0, u) \sim (u, 0))$. The images $d, t: \square^1 \rightarrow W_1$ of the diagonal and the top edge are levelwise injective, and their images meet exactly in the degenerate cells of the common vertex v_{11} ; hence the wedge $d \vee_{v_{11}} t \rightarrow W_1$ is a monomorphism.*

Proof. (i) is immediate: distinct cells of a representable. (ii) The congruence of W_1 is generated levelwise by the pairs $(0, w) \sim (w, 0)$, $w \in D(k)$; since no cell of the form $(0, w)$ or $(w, 0)$ equals another such cell except as forced, every congruence class is either a singleton or one swap pair $\{(0, w), (w, 0)\}$. A diagonal cell (u, u) lies in a swap pair only if $u = 0$, i.e. only the degenerate cells of v_{00} ; a top cell $(u, 1)$ only if $u = 1$ hits nothing (the pair $(0, w)$ requires a zero first component, $w = 1$ gives $(0, 1) \neq (u, 1)$ unless $u = 0$, and $(w, 0)$ has second component $0 \neq 1$); injectivity of d and t and the computation of $d \cap t$ follow by inspection of these classes: $(u_1, u_1) \sim (u_2, u_2)$ forces $u_1 = u_2$, $(u_1, 1) \sim (u_2, 1)$ forces $u_1 = u_2$, and $(u, u) \sim (w, 1)$ forces $u = w = 1$. (These facts are also machine-verified through level 3; artifact.) $\square$

Proposition 4.2. $W_{\wedge\vee}$ is test-contractible.

Proof. Identifying two parallel cells $A \sim B$ that share their source vertex is the pushout of the fold map along the wedge inclusion:

$$\square^2 / (A \sim B) = \square^2 \cup_{A \vee B} \square^1,$$

where $A \vee B$ is the wedge at the shared vertex, included by (A, B) , and the right leg folds both intervals onto one. For the swap fold, Lemma 4.1(i) makes the wedge inclusion a monomorphism, so the square is a homotopy pushout of test-contractible objects ($\square^2$, the wedge, and $\square^1$ are contractible), and $W_1 \simeq_{\text{test}} *$. In W_1 the diagonal and top edges share their target vertex v_{11} (and, after the first fold, $d(0) = v_{00} \neq v_{01} = t(0)$), and Lemma 4.1(ii) makes the second wedge inclusion a monomorphism, so

$$W_{\wedge\vee} = W_1 \cup_{d \vee t} \square^1$$

is again a homotopy pushout of contractibles, and $W_{\wedge\vee} \simeq_{\text{test}} *$. $\square$

Proof of Theorems 1.1 and 1.3. $W_{\wedge\vee} \rightarrow *$ is a test weak equivalence by Proposition 4.2. It is not a cubical-type weak equivalence: otherwise ι would be homotopic to a constant (Lemma 2.3), contradicting Theorem 3.2. $\square$

5. MACHINE VERIFICATION

All finite claims are verified in the artifact of the series (release accompanying this paper, version-pinned with a Zenodo DOI): the base facts of Lemma 3.1 by two independent implementations (a congruence-closure engine over all site maps, and a generator-only coequalizer formulation with a different closure algorithm), agreeing on all counts (level sizes 2, 5, 26, 362; seven endomorphism classes; nine cylinders with an identity end among old cells, of which exactly the constant is valid); Lemma 1.4 over all 20 elements of $D(3)$; the strict contractibility of both single-fold quotients; the wedge monomorphism facts of Lemma 4.1 through level 3; the two-fold blocking of both connection cones; and exhaustive fresh-cylinder searches over filling stages with boxes of dimension ≤ 4 (462,672 candidates at dimension 4), all empty, as the theorem predicts.

Data availability. Scripts and the pinned artifact are archived with the series; the release accompanying this paper is `v1.21.0`, DOI 10.5281/zenodo.22038185.

Use of AI tools. An AI assistant was used in the preparation of this manuscript and its verification scripts; the author takes full responsibility for the content.

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